MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, XM712, WARHEAD FOR

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for the manufacture and acceptance of the Warhead, Guided Projectile XM712.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Box, Wood, Cleated Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-850	-	Cushioning Material, Polystyrene Expanded, Resilient (For Packaging Application)
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular, Polyethylene Foam, Flexible
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)

MILITARY

MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaged, Use and Static Dehumidification
MIL-S-14760	-	Silica
MIL-P-17667	-	Paper, Wrapping, Chemically Neutral (Noncorrosive)

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	-	Marking For Shipment And Storage
MIL-STD-410	-	Nondestructive Testing, Personnel Qualification And Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic, and Ultrasonic)
MIL-STD-453	-	Inspection, Radiographic
MIL-STD-810	-	Environmental Test Methods

DRAWINGS

9300922	-	Adhesive, Explosive Compatible Epoxy, Two Component
9300967	-	Fabrication of Control Section and Warhead Body Housing, Process Specification for
9302004	-	Liner, Shaped Charge-Precision, 42 Degrees
9307000	-	Warhead, Guided Projectile, High Explosive
9307001	-	Body, Warhead
9307005	-	Retainer Ring, Liner
9307006	-	Cone, Warhead Free Space
9307008	-	End Closure, Warhead
9307009	-	Seal, End Closure
9307011	-	Warhead, Empty, Guided Projectile
9307025	-	Retainer, Fuze
9307046	-	Wiring Harness, Warhead - 3W1 - 3W2
9307051	-	Aft Closure, Warhead Forging
9307055	-	Warhead Loaded, Guided Projectile
9307058	-	Warhead Acceptance Penetration Criteria (U)

OTHER PUBLICATIONS

REGULATIONS

AFR-71-4	-	Preparation of Hazardous Material for Military Air Shipment
AR-700-15	-	Packaging of Material

EQUIPMENT OPERATING INSTRUCTIONS

EOI 9304456	-	Warhead Test Set, Equipment Operating Instruction

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- The Warhead, Guided Projectile XM712, (drawing 9307000), hereinafter referred to as the warhead section, includes a shaped-charge warhead, fuze module, retainer, seal, end closure, and wiring harnesses which are contained within a tubular-shaped steel body. The outer surface of the body forms the outer surface of the Guided Projectile XM712, hereinafter referred to as the projectile. The electrical harnesses are routed through and bonded to the inner surface of the warhead section body. An "empty warhead assembly" is a completely assembled warhead section less the explosive charge, fuze module, retainer, seal, and end closure. A "loaded warhead assembly" is a completely assembled warhead section, including the explosive charge, less the fuze module, retainer, seal, and end closure.

3.1.1 Government furnished property.- Each warhead section will contain high explosive, furnished and loaded by the Government.

3.2 Characteristics.- Characteristics of the empty warhead assembly, loaded warhead assembly, and warhead section shall be as specified herein.

3.2.1 Performance

3.2.1.1 Empty warhead assembly wiring harness.- The electrical wiring contained in the empty warhead assembly shall withstand a 500 ±25 volt direct current (vdc) insulation resistance test of each wire within a cable to each adjacent wire within the same cable, and from each wire within a cable to the warhead section metal body. The minimum insulation resistance under these conditions shall be 1000 megohms. The end-to-end resistance of each wire within the wiring harness shall be as specified on drawing 9307046.

3.2.1.2 Loaded warhead assembly wiring harness.- The electrical wiring contained in the loaded warhead assembly shall withstand one milliamp max direct current (dc) insulation resistance test of each wire within a cable to each adjacent wire within the same cable, and from each wire within a cable to the warhead section metal body. The minimum insulation resistance under these conditions shall be greater than 20 megohms. The end-to-end dc resistance of each wire within the wiring harness shall be as specified on drawing 9307046. This requirement does not include fuze wiring.

3.2.1.3 Warhead section wiring.- The electrical characteristics of the warhead section wiring after installation of the fuze module in the loaded warhead shall be in accordance with Table I.

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3.2.1.4 Leakage.- The leakage rate of the empty warhead charge cavity, including the aft wire slots, shall not exceed 10^{-4} cubic centimeters per second (cc/sec) when exposed to an internal gas pressure of 15 ± 1.0 pounds per square inch gage (psig).

3.2.1.5 Cracks and cavities

3.2.1.5.1 Cavity dimensions.- Cavities having a maximum projected dimension of 1/32 inch or less shall be disregarded, provided they are separated by 1/2 inch or more of solid explosive. The maximum sum of the projected areas of cavities, and the maximum dimension of any cavity shall be as specified in Table II for each segment defined in Figure 1. A projected dimension of a cavity in the explosive charge is defined as a dimension that can be measured on a radiograph of the loaded warhead.

3.2.1.5.2 Cavities extending into two segments.- The projected area of cavities extending into two segments shall meet the requirements of both segments.

3.2.1.5.3 Cracks and annular rings.- Cracks having a projected width not greater than 1/32 inch shall be disregarded. The presence of annular rings 1/16 inch maximum projected thickness and extending from the steel wall 1/4 inch into the charge shall not be cause for rejection. The projected area represented by such a crack or annular ring shall not be included when determining cavity area for the charge. The maximum number of cracks allowed shall be in accordance with Table II.

3.2.1.5.4 Piping cavity.- A piping cavity on the longitudinal axis of the projectile between the apex of the liner and the faced surface of the charge shall be permitted if its projected length does not exceed 1 1/2 inches or its width does not exceed 1/8 inch. The projected area of such a cavity shall not be included when determining the cavity area of the charge. The faced surface of the explosive charge includes the bottom and sidewalls of the fuze well.

3.2.1.5.5 Visible cavities.- The maximum dimensions of any visible cavity in the machined surface of the explosive charge shall not exceed 3/16 inch. Individual cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the fuze well circumference. The total area of all cavities in the base of the fuze well shall not exceed 1/4 square inch. Cavities in the machined surface are defined as discontinuities. Only that portion of such discontinuities deeper than 0.030 inch shall be included in the total area.

3.2.1.5.6 Base separation.- At any location around the circumference of the aft opening, an 0.030 inch thick by 1/4 inch wide feeler gage shall not enter more than 0.250 inch between the explosive and the warhead body. No base separation over 0.030 inch between the angled portion of the body

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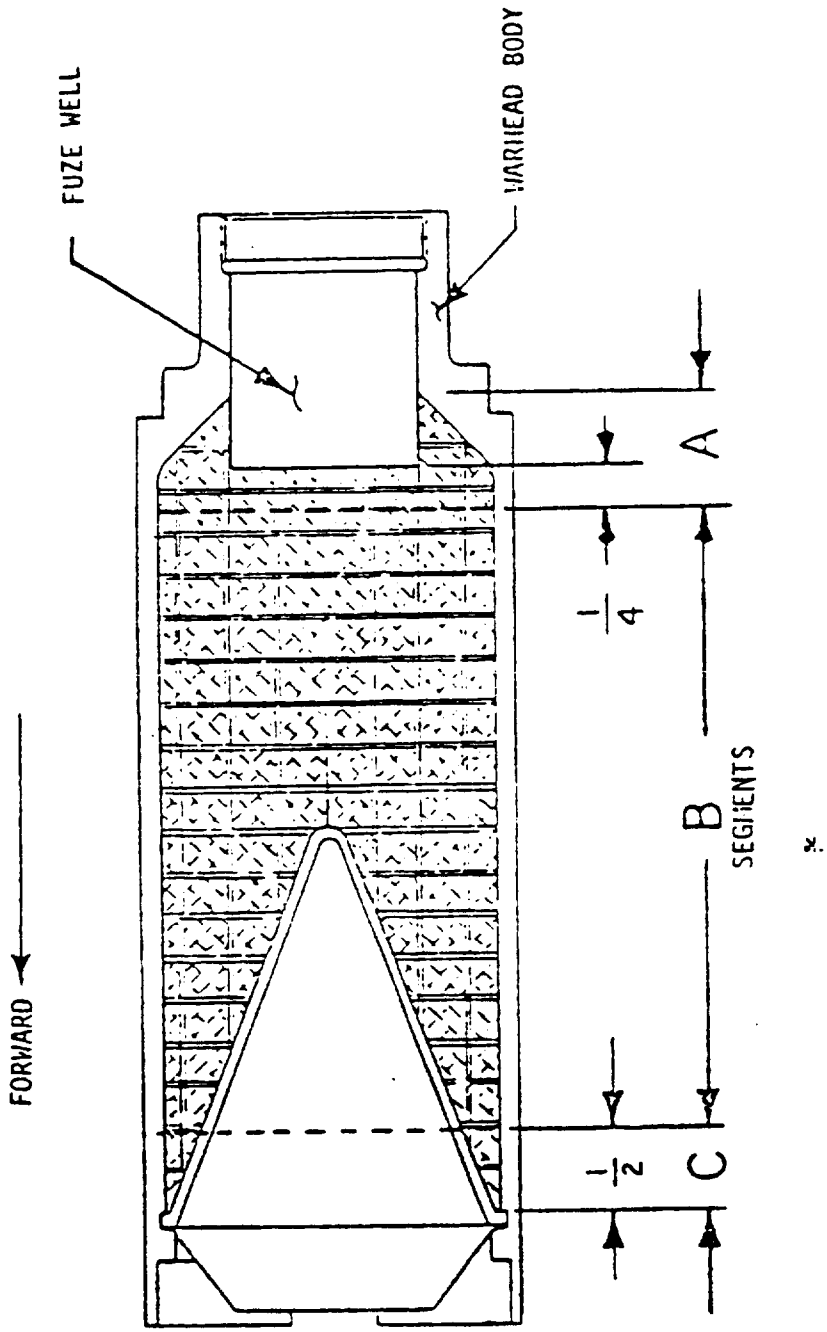


Figure 1. Division of Explosive Charge into Segments for Radiographic Examination of Cavitation - Dimensions in Inches

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inside diameter and the aft end of the explosive charge shall be permitted as indicated on radiographs of the charge.

3.2.1.5.7 Classification of defects

3.2.1.5.7.1 Critical defects.- Warheads which contain explosive having the following defects shall be classified as critical defectives and are not acceptable for gun firing.

(a) Cavities, cracks or annular rings, in segments A or B (Figure 1), in excess of twice that permitted by the applicable requirement.

(b) Evidence of explosive leakage.

(c) Base separation in excess of that permitted by the applicable requirement.

3.2.1.5.7.2 Major defects.- Warheads which contain explosive having the following defects shall be classified as major defectives and shall be submitted to the procuring activity for disposition.

(a) Three or more cracks having a combined width in excess of 1/16 inch crossing the same line drawn parallel to the longitudinal axis of the warhead in segment A.

(b) Cavitation, cracks or annular rings in segment B in excess of that permitted by the applicable requirement, but not in excess of twice that permitted.

(c) Cavitation, cracks or annular rings in segment C in excess of that permitted by the applicable requirement.

(d) Piping cavity in excess of the applicable requirement.

(e) Surface cavities in excess of that permitted by the applicable requirement.

3.2.1.6 Explosive leakage.- There shall be no evidence of explosive material forward of the warhead body/shaped-charge liner joint after pouring of the explosive material into the explosive charge cavity.

3.2.1.7 Warhead cavity finish/cleanliness.- Prior to loading there shall be no foreign matter, no unsolidified paint, and no unpainted (unprimed) steel surfaces in the explosive cavity. The interior surface of the cavity shall be clean and dry when the explosive charge is loaded. After loading there shall be no explosive in any location of the loaded warhead other than in the explosive cavity as defined by drawing 9307055. There shall be no loose explosive material left in the fuze well after the machining operation has been completed.

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3.2.1.8 Static penetration.- The warhead static penetration criteria are specified by drawing 9307058 (Confidential).

3.2.2 Environmental conditions.- After exposure to the following environmental conditions the loaded warhead assembly shall show no evidence of physical damage such as explosive leakage, cracking, or degradation which could in any manner prevent the warhead from meeting the requirements of 3.2.1.

3.2.2.1 Storage and transportation temperatures.- The loaded warhead assembly shall withstand exposure to ambient temperatures from 160°F (71°C) through -50°F (-45.6°C) and temperature shock in either direction between these extremes.

3.2.2.2 Transportation vibration.- The loaded warhead assembly shall withstand exposure to the transportation vibration requirements of MIL-STD-810, Method 514.2, Procedure X.

3.2.2.3 Fragility shock.- The loaded warhead assembly shall withstand exposure to half sine wave mechanical shock pulses having nominal durations up to and including 10 milliseconds (ms) with nominal peak accelerations of 200g.

3.2.2.4 Launch shock.- The loaded warhead assembly shall withstand a forward longitudinal acceleration of 9000g ± 450g at temperatures ranging from -25°F (-31.7°C) through 145°F (63°C).

3.3 Design and construction

3.3.1 Production drawings.- The warhead section shall be fabricated and assembled in accordance with drawings, parts lists, and other documents listed on drawing 9307000.

3.3.2 Workmanship.- All details of workmanship shall be of the highest grade consistent with this specification. There shall be no cracks, dents, scratches, burrs, sharp edges, loose parts, foreign matter, or other evidence of poor workmanship that will render the warhead section unsuitable for its intended use. Cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by any cleaning agents.

3.4 First article inspection.- This specification makes provision for first article inspection. The first article shall meet all requirements of this specification. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the

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performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The following types of inspections shall be conducted on this item:

(a) First article inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 First article inspection

4.3.1 Submission.- The contractor, when required by contract, shall submit a first article sample of 10 empty warhead assemblies for examination and loading as defined by Table III. The contractor shall additionally submit 5 complete sets of parts and assemblies as listed in "Classification of Defects and Tests," Table V, sheets 1 through 13. The ten empty warhead assemblies shall have been inspected in accordance with Table V prior to loading. After loading, the ten loaded warhead assemblies shall be inspected in accordance with Table IV. Environmental tests shall be conducted in the sequence indicated. The operating performance of each warhead shall be as indicated in each test method. The testing shall be considered destructive. All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and material shall be obtained from the same source(s) of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the procuring activity. A first article sample, or portion thereof, as directed by the procuring activity, shall also be submitted whenever there is a lapse in production for a period in excess of 180 days, or whenever a change occurs in manufacturing process, material used, drawing or specification, such as to significantly affect the product uniformity as determined by the procuring activity.

4.3.2 Inspection to be performed.- First article test specimens may be subjected by the procuring activity to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings.

4.3.3 First article rejection.- If any test specimen fails to comply with any of the applicable requirements, the first article samples shall

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be rejected. The procuring activity reserves the right to terminate its inspection upon failure of any test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection

4.4.1 Inspection lot.- Unless otherwise specified herein or by the procuring activity, inspection lot size, lot formation, and presentation of lots shall be in accordance with the inspection and sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be not greater than 1 month's procuction. Each lot of warhead sections shall consist of warhead body forgings made from a single heat of steel and tubes made from a single heat of steel. The fuze modules shall be from one lot. Composition B shall be of the same grade from one manufacturer from one interfix.

4.4.2 Inspections

4.4.2.1 Sampling inspection.- Each lot of warhead sections shall consist of hardware which has been inspected per Table V. Unless otherwise specified, sampling shall be to level II of MIL-STD-105.

4.4.2.2 Lot acceptance.- Lot acceptance shall be based on acceptance of individual sublots. From each sublot, consisting of warheads poured from explosive melted at one time in a single melt kettle, two units shall be selected at random for static penetration in accordance with 4.5.8. Failure of any loaded warhead assembly to meet the requirements of drawing 9307058 (Confidential) shall be cause for rejection of the sublot and processing in accordance with 4.4.3.2. Sublot acceptance tests shall be performed in accordance with Table VI.

4.4.3 Conformance inspection rejection

4.4.3.1 Sampling inspection rejection.- Lots failing to meet the requirements of 4.4.2.1 shall be dispositioned in accordance with the applicable requirements of MIL-STD-105.

4.4.3.2 Sublot rejection.- Any sublot failing to meet the requirements of 4.4.2.2 shall be resubmitted once by selecting two additional loaded warhead assemblies for test. This test set-up shall make provisions for flash X-ray to record radiographs of the shaped charge jet in two planes at 90 degrees apart. If both loaded warhead assemblies meet the requirements of drawing 9307058 (Confidential), the sublot shall be accepted. Failure to meet the requirements of drawing 9307058 (Confidential) shall cause the sublot to be rejected. Rejected sublots, with test data and radiographs, shall be submitted to the procuring activity for disposition.

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4.5 Test methods.

4.5.1 Test conditions

4.5.1.1 Environmental conditions.- Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

(a)	Temperature:	72° ± 18°F (22.2° ± 10°C)
(b)	Relative Humidity:	90 percent maximum
(c)	Barometric Pressure:	28 to 32 inches of mercury

4.5.1.2 Test tolerances.- All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

4.5.1.3 Inspection equipment.- The inspection equipments required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the inspection tables included herein. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (See 6.4).

4.5.2 Physical and dimensional characteristics inspection.- The physical and dimensional characteristics shall be inspected to the applicable requirements of drawing 9307000.

4.5.3 Leak test.- The empty warhead assembly shall be pressurized to 15 ± 1.0 psig. The leakage rate shall not exceed the equivalent of 10^{-4} cc/sec of air at standard conditions.

4.5.4 Electrical characteristics test

4.5.4.1 Empty warhead assembly test.- The electrical characteristics of the wiring harness shall be tested to the requirements of 3.2.1.1.

4.5.4.2 Loaded warhead test.- The electrical characteristics of the wiring harness shall be tested to the requirements of 3.2.1.2.

4.5.4.3 Warhead section test.- The electrical characteristics of the warhead section shall be tested in accordance with EOI 9304456 and shall meet the requirements of Table I. Resistance measurements shall be made with the positive output of the meter connected to the "From" conductor and the negative output of the meter connected to the "To" conductor of Table I. The test current shall be limited to 1 milliamp or less.

4.5.5 Explosive charge inspection.- Two radiographs of each loaded warhead assembly shall be taken at 90 degrees apart and normal to the

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longitudinal axis. The source-to-film distance and radiation energy shall be as specified in 4.5.5.2. Projected dimensions as seen on the radiographs are to be compared to Table II, as defined by 3.2.1.5.1 through 3.2.1.5.4. Defects in excess of those defined shall be categorized per 3.2.1.5.7.

4.5.5.1 Radiographic exposures and examination.- Warheads shall be subjected to radiographic inspection on a 100 percent basis after loading and fuze well machining. An approved type of radiographic film (4.5.5.8) shall be used in conjunction with rigid metal cassettes. Lead, tantalum, or tungsten metal screens of appropriate thickness and finish shall be used in contact with all points of the film during exposures. Cassettes and screens shall be inspected periodically, and shall be replaced upon evidence of wear or damage that might degrade radiographic quality. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features shall appear on the area of radiographs under inspection, with the exception of specified penetrameters (4.5.5.4) and approved radiation shielding used to prevent over-exposure of film in selected areas. Such shielding, if used, shall be free of any radiographically observable features in the projected area under inspection. Radiographic exposures and film processing shall be controlled to consistently meet film density requirements of 4.5.5.5. Examination of radiographic images shall include the treatment of all potential defect features as actual defects, unless positively identified as extraneous artifacts or acceptable metal parts irregularities. Additional radiographs of an item may be taken to improve visibility or to properly classify a defect. All such additional radiographs shall be identified as retakes and traceable to the original radiographs.

4.5.5.2 Radiography of warheads.- Warheads shall be positioned for radiographic exposures to produce no more than two warhead images on one 14 inch by 17 inch film. The length of one warhead shall be covered by one film extending from the flange on the shaped-charge cone back to the threads for the fuze retainer. The complete elliptical projection of the flange on the cone shall be visible on the radiograph. The radiation source shall lie in a plane perpendicular to the longitudinal axis of the projectile, such that no discernible parallax is produced on the radiographic image at the position of the rear surface intersecting the largest outside diameter. Source-to-film distance shall be not less than 70 inches. The peak radiation energy shall be equal to or greater than 900 kilovolts. Each warhead shall be radiographed in two orientations, separated by a rotation of 90 degrees about its longitudinal axis, and aligned on consistent external features. Penetrameters and other radiographic sensitivity indicators shall be employed as specified in 4.5.5.4. Radiographs of loaded warheads shall be examined by personnel certified in accordance with MIL-STD-410 for all sensitivity indicators and radiographic inspection requirements of this specification. Warheads which prove defective in one or both orientations shall be rejected.

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4.5.5.3 Radiographic identification.- An identification numbering system shall be employed which shall provide direct, precise correlation between each item and its radiographic image, and which shall be used for reporting purposes.

4.5.5.4 Radiographic sensitivity.- The sensitivity of the radiographic technique for warhead inspection shall be determined by means of penetrameters fabricated in accordance with MIL-STD-453 and as specified herein, and shall be positioned on the radiographic image as shown in Figure 2. Two penetrameters for each warhead shall be used, and the positions of the 2T holes shall be closely controlled; with the penetrameters on the radiation source side of the projectile. Penetrameters shall be made of steel having an approximate specific gravity of 7.8. When radiographing a number of similar warheads, the frequency of sensitivity check shall be once each radiographic exposure. One warhead may be used repeatedly for the penetrameter exposures. For in-motion exposures on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring. For stationary exposures of a number of warheads, the penetrameters shall be used on the warhead farthest from the central beam of radiation. The radiographic materials and technique used on the warhead having penetrameters shall be identical in every controllable aspect to those used on all other inspected warheads. The sensitivity of the radiographic technique shall not be considered satisfactory unless the outline of the penetrameters and the image of the 2T holes are perceptible on the radiographs. Failure of this sensitivity check shall be cause for cessation of radiographic operations until the source of the failure is corrected. All warheads radiographically accepted subsequent to the last successful sensitivity check shall be re-radiographed and reinspected. Additionally, the sensitivity of the radiographic technique for warhead inspection shall be determined by examination of internal features. These features shall consist of: (1) the complete outline of the internal surface of the steel warhead body surrounding the explosive and fuze well; (2) the complete outline of both the inner and outer surfaces of the shaped-charge cone; (3) the complete outline of the fuze well in the explosive. The sensitivity of the radiographic technique shall not be considered satisfactory unless these features are sharply defined on the radiographic image of every warhead inspected. Failure to meet those sensitivity requirements on any warhead shall require re-radiographing of the warhead and a diagnosis of the failure.

4.5.5.5 Photographic density of radiographic film.- The photographic density of the film of warheads shall be determined in the region of minimum film density within the projected area of the explosive charge, and shall measure 1.25 or greater. Film density shall be checked each hour of operation or fraction thereof for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units and having a sample spot of 3 millimeters diameter. Accuracy of densitometer shall be maintained by

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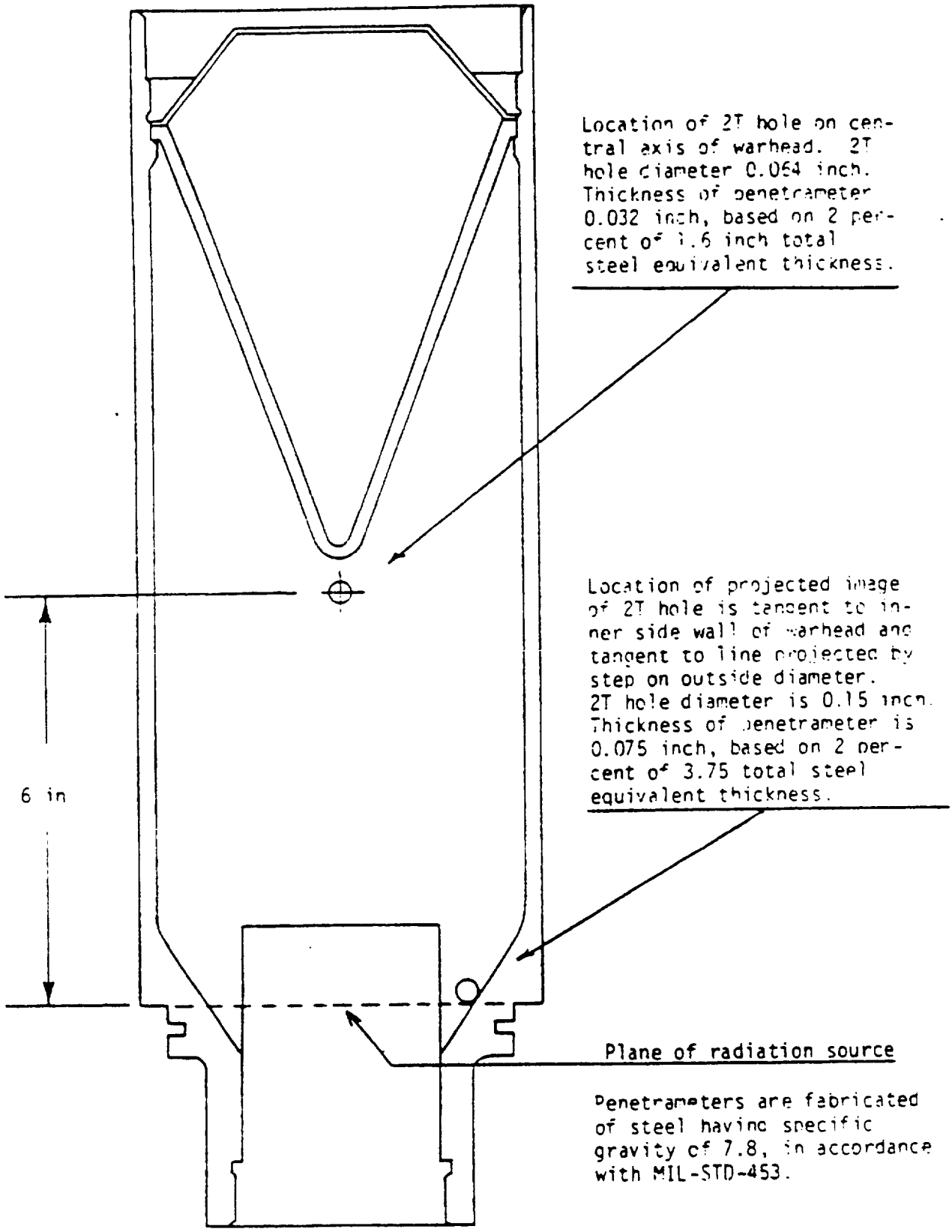


Figure 2 - Penetrameter Locations

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calibrating with a photographic step tablet each hour immediately prior to measurements of the radiographic film. Calibration shall be made at a step tablet density level within .2D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablets shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet used in daily operations having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.5.6 Radiographic procedures.- Radiographic inspections shall be performed in accordance with the general requirements of MIL-STD-453 and specific product assurance procedures prepared by the contractor and approved by the Government procuring activity.

4.5.5.7 Review of radiographic sensitivity and techniques.- A radiograph of warheads showing penetrameters, as well as radiographic diagrams and techniques, shall be submitted for review and approval prior to the start of initial production. The initial radiograph will be retained at ARRADCOM. One radiograph of warheads showing penetrameters shall be submitted for review once each month of production on each contract. All radiographs submitted shall be produced using materials and techniques identical to those used on normal production items.

4.5.5.8 Radiographic film.- Only approved industrial radiographic film shall be used to radiograph the loaded warheads or components thereof. Types, other than those listed below, may be used only after evaluation and approval by ARRADCOM. The following are initially approved types:

Eastman Kodak Company, Types AA, T, and M.
E.I. Dupont De Nemours & Co., Types NDT-55, NDT-65, and NDT-70.
GAF Types 400 and 800.

4.5.5.9 Nonfilm radiographic inspection.- No fluoroscopic or other nonfilm method of radiographic imaging shall be substituted for film radiography prior to specific review and approval by the procuring activity. In general, any such method must meet all radiographic image quality and sensitivity requirements of this specification, as well as other required characteristics inherent to film imaging.

4.5.6 Fuze well inspection.- Visual inspection of the fuze well after machining shall show no cavities larger than those specified in 3.2.1.5.5. There shall be no base separation when probed with a 0.03C

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inch thick by 1/4 inch wide feeler gage, as specified in 3.2.1.5.6. Defects in excess of those defined shall be categorized per 3.2.1.5.7.

4.5.7 Environmental tests.- Environmental tests shall be performed on the loaded warhead assembly. The loaded warhead assembly shall be prepared for environmental testing by installation of a test fuze (See 6.3.3) to be provided by the test agency.

4.5.7.1 Storage/transportation temperature test.- Thermal environmental testing shall be accomplished in accordance with MIL-STD-810, Method 503.1, Procedure I, with the following exceptions: the low temperature shall be -50°F (-45.6°C), the first high temperature period shall be a minimum of 48 hours, the last low temperature period shall be a minimum of 24 hours, and the unit under test (UUT) stabilization time at the intermediate high/low temperature points shall be 8 hours minimum.

4.5.7.2 Transportation vibration test.- The UUT shall be tested for vibration in accordance with MIL-STD-810, Test Method 514.2, Figure 514.2-7, Curve AX, Table 514.2-VII except search and dwell tests shall be omitted. Adaption of the UUT to the vibration machine interface shall include support at the normal fore and aft points of attachment of the UUT.

4.5.7.3 Fragility shock test.- The UUT shall be tested for fragility shock in accordance with MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2, with the exception that the peak value (A) shall be 200 g's and the nominal duration (D) shall be 10 milliseconds.

4.5.7.4 Launch shock test.- Launch shock tests shall be performed on the loaded warhead assembly. Each UUT shall be separately placed in a canister and subjected to one of the temperature environments specified in 4.5.7.4.1 and 4.5.7.4.2. Within 30 minutes after being removed from the temperature conditioning environment, the canister assembled UUT shall be subjected to the acceleration requirement of 3.2.2.4 by launching from an 8 inch gun, M115 or equivalent (GFE).

4.5.7.4.1 High temperature test.- The UUT shall be placed in a high temperature chamber at 145° ± 5°F (62.8° ± 3°C) and allowed to remain for a minimum of 16 hours prior to testing in accordance with 4.5.7.4.

4.5.7.4.2 Low temperature test.- The UUT shall be placed in a low temperature chamber at -25° ± 5°F (-31.7° ± 3°C) and allowed to remain for a minimum of 16 hours prior to testing in accordance with 4.5.7.4.

4.5.8 Static penetration test.- The UUT shall be tested for static penetration in accordance with the requirements and procedures specified in drawing 9307058 (Confidential). Test fuze is defined in 6.3.3.

4.5.9 Warhead cavity inspection.- The warhead cavity shall be inspected to verify conformance to the finish/cleanliness requirements of 3.2.1.7.

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4.5.10 Workmanship inspection.- The warhead section, parts, and assemblies shall be visually inspected to verify conformance to the requirements of 3.3.2

4.5.11 Preparation for delivery inspection.- The warhead section shall be visually inspected to verify conformance to the packaging and packing requirements of the procurement document (see 6.2(f)).

5. PREPARATION FOR DELIVERY

5.1 General.- Packaging and packing, shall be level A, B, or commercial, as specified (see 6.2(f)).

5.1.1 Level A

5.1.1.1 Cleaning.- Cleaning shall be accomplished by wiping the exterior surfaces of the warhead assembly with a clean, dry, lint-free cloth. The internal portions of the assembly may be cleaned with clean, dry, compressed air.

5.1.1.2 Packaging.- The warhead assembly shall be packaged in accordance with AR700-15 Level B. Cables shall be protected against damage. Sharp edges and protrusions shall be covered with PPP-C-843 cushioning material. The assembly shall then be wrapped with Type II, MIL-P-17667 barrier material. Desiccant conforming to MIL-D-3464 shall be spaced uniformly throughout the assembly and secured with PPP-T-76 tape. The assembly shall then be heat sealed within a class F, MIL-B-117 bag.

5.1.1.3 Packing.- The packaged warhead shall be cushioned within the exterior container with polystyrene or polyethylene end pad and saddles, PPP-C-850, or PPP-C-1752, respectively. The exterior container shall conform to PPP-B-601 or PPP-B-621, overseas requirements.

5.1.2 Level B

5.1.2.1 Cleaning.- See 5.1.1.1.

5.1.2.2 Packaging.- The warhead assembly shall be packaged in accordance with AR700-15, Level B. Cables shall be protected against damage. Sharp edges and protrusions shall be cushioned with PPP-C-843. The assembly shall then be heat sealed within a class F, MIL-B-117 bag.

5.1.2.3 Packing.- Packing shall be accomplished as defined in 5.1.1.3 with the exception that the exterior container shall be a domestic type.

5.1.3 Commercial

5.1.3.1 Cleaning.- See 5.1.1.1.

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5.1.3.2 Packaging.- The warhead assembly shall be packaged in a manner that will prevent contamination of the item and damage to cables, during shipping from the supply source to the first receiving activity for immediate use.

5.1.3.3 Packing.- The warhead assembly, either single unit, or multiples thereof, shall be properly secured in a strong wooden or metal box. The gross weight of a box containing one or more assemblies shall not exceed 300 pounds.

5.2 Marking.- In addition to the standard unit and exterior markings required in MIL-STD-129, each exterior container shall be marked "EXPLOSIVE PROJECTILE" and have affixed thereto a class A explosive label.

5.3 Transportation.- Explosive projectiles are not acceptable for commercial air shipments. However, they are acceptable for Military Air Shipments, in accordance with Air Force Regulation 71-4.

6. NOTES

6.1 Intended use.- The warhead assembly is intended for use as a section of the XM712 Projectile.

6.2 Ordering data.- Procurement documents shall specify the following:

6.2.1 Procurement requirements

(a) Title, number and date of this specification.

(b) Quantity required and delivery schedules.

(c) Serialization requirements, if applicable.

(d) Quality Conformance Inspection, if other than specified in Section 4 of this specification.

(e) First Article Sample requirements, if other than specified in Section 4 of this specification.

(f) Packaging requirements, if other than specified in Section 5 of this specifiction.

(g) Certificates of conformance for each lot or shipment of product.

(h) Requirements for lot control.

6.2.2 Contract data requirements

(a) Contract data requirements for inspection equipment designs (conforming to Data Item Description DI-E-1118 or DI-R-1714 with appropriate addendum to each).

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(b) Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions

6.3.1 Contractor.- Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commerical contractors, sub-contractors, Government Owned, Contractor Operated (GOCO) plants.

6.3.2 Technical data package (TDP).- A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists of specifications, standards, models, performance requirements, quality assurance provisions, and packaging data.

6.3.3 Test fuze.- The test fuze used for static firing of warheads shall consist of an aluminum body having an outside diameter of $3.035 \pm \begin{smallmatrix}0\\0.010\end{smallmatrix}$ inches and a length of 3.50 ± 0.03 inches. The forward end shall have a chamfer of 45° x 0.17 inch on its diameter and a central recess containing a booster charge 0.50 ± 0.03 inch diameter by 0.50 long of PBXN-5 explosive. The body shall have an axial through hole of diameter suitable to contain a detonator to be selected by the test agency. Installation of the test fuze in the warhead shall include means for retaining the fuze firmly in contact with the warhead main charge and for insuring firm contact between the detonator, booster, and main charge.

6.4 Inspection equipment design.- Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs.- SIE designs may consist of any of the following:

(a) Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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(b) A source control drawing or a specification control drawing as defined in MIL-STD-100.

(c) An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor design.- Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commerical equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

6.4.3 Submission of designs for approval.- Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract.

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Table I.

WARHEAD SECTION RESISTANCE

Conductor Number		Function	Resistance
From	To		
1	5	Fire Detonator to Fuze Ground	320K ohms to 400K ohms
5	1	Fuze Ground to Fire Detonator	50K ohms to 90K ohms
2	5	Explosive Actuator Squib Drive to Fuze Ground	4.0 to 12.2 ohms
3	5	Squib Ground to Fuze Ground	0 to 1 ohms
4	5	Charge Detonator Capacitor to Fuze Ground	900K ohms minimum after 30 seconds
1, 2, 3, 4, 5	Case	Wire to Case	Open Circuit (>20 megohms)
6 thru 18		Continuity (End-to-End)	as specified on drawing 9307046
6 thru 18		Wire to Wire	Open Circuit (>20 megohms)
6 thru 18		Wire to Warhead Body	Open Circuit (>20 megohms)

Table II

CRACK AND CAVITY CRITERIA

Characteristic	Segment		
	A	B	C
Sum of Projected Areas of Cavities (sq inch)	1/8	3/8	1/4
Projected Dimensions of Any Cavity (inch)	3/8	5/8	1/2
Maximum Number of Cracks Allowed	4	3	2

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Table III.

DISTRIBUTION AND SAMPLE SIZE FOR FIRST ARTICLE INSPECTION

	Qty	Location of Tests & Examination	Identification of Tests and Examinations	Disposition of Samples
Empty Warhead Assembly (DWG. 9307011)	10	Manufacturing and Assembly Plant	See Table V	Continue to Next Assembly Level (Dwg. 9307055) to be Tested per Table IV
Loaded Warhead Assembly (Dwg. 9307055)	5	Ammunition Loading Plant	See Table IV, Section A	Parts Expended in Test
	5	Assembly Plant	See Table IV, Section B	Parts Expended in Test
Parts/Assemblies	5 Sets	Manufacturing and Assembly Plant	See Table V	As Directed by Procuring Activity

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Table IV - SECTION A
FIRST ARTICLE INSPECTIONS, LOADED WARHEAD ASSEMBLY
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE WARHEAD, LOADED, GUIDED PROJECTILE (10 ASSEMBLIES AT AMMUNITION LOADING PLANT)	SHEET 1 OF 3		DRAWING NUMBER 9307055 NEXT HIGHER ASSEMBLY 9307000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	No explosive material, other than within charge cavity.	100%	3.2.1.6	4.5.9
2	No loose explosive material in the fuze well	100%	3.2.1.7	4.5.9
3	Defects in explosive charge	100%	3.2.1.5.1	4.5.5, 4.5.6, 4.5.9
Major				
101	Loading procedure	100%	Dwg	Gage/Visual
102	Specific gravity of explosive	Lot Sample	Dwg	Gage
103	Defects in explosive charge	100%	3.2.1.5.2	4.5.5
104	Static penetration (5 assemblies only, see Note 1)	100%	3.2.1.8	4.5.8

NOTES: 1 - Test firing of 5 units only. Remaining units to be returned to assembly plant for environmental tests. Test fuze defined in 6.3.3.

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Table IV - SECTION A
FIRST ARTICLE INSPECTIONS, LOADED WARHEAD ASSEMBLY
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE WARHEAD, LOADED, GUIDED PROJECTILE (10 ASSEMBLIES AT AMMUNITION LOADING PLANT)	SHEET 2 OF 3		DRAWING NUMBER 9307055 NEXT HIGHER ASSEMBLY 9307000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Minor				
201	Workmanship	100%	3.3.2	4.5.10
202	Warhead lot number	100%	Dwg	Visual
203	Electrical harness inspection for damage	100%	3.3.2	4.5.10
204	Fuze well depth	100%	Dwg	Gage
205	Fuze well diameter	100%	Dwg	Gage
206	Location of fuze well within metal body ID	100%	Dwg	Gage

NOTES:

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Table IV - SECTION B
FIRST ARTICLE INSPECTIONS, LOADED WARHEAD ASSEMBLY
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH CATEGORY	TITLE WARHEAD, LOADED, GUIDED PROJECTILE (5 ASSEMBLIES AT ASSEMBLY PLANT - AFTER LOADING - SEE NOTES 2 & 3)	SHEET 3 OF 3		DRAWING NUMBER 9307055 / NEXT HIGHER ASSEMBLY 9307000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None Defined			
Major				
101	Wiring harness electrical characteristics	100%	3.2.1.2	4.5.4.2
102*	Storage/Transportation Temperature	100%	3.2.2.1	4.5.7.1
103*	Transportation Vibration	100%	3.2.2.2	4.5.7.2
104*	Fragility shock	100%	3.2.2.3	4.5.7.3
105*	Launch shock (see Note 4)	100%	3.2.2.4	4.5.7.4
106*	Wiring harness electrical characteristics	100%	3.2.1.2	4.5.4.2
107*	Static Penetration	100%	3.2.1.8	4.5.8
Minor				
201	Workmanship	100%	3.3.2	4.5.10

NOTES: 2 - Test fuze in accordance with 6.3.3 to be supplied by contractor and installed prior to tests marked*.
3 - No deviations from sequential order of tests permitted.
4 - Two assemblies at +145°F and three assemblies at -25°F.

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: LINER	SHEET 1 OF 16		DRAWING NUMBER: 9302004
				NEXT HIGHER ASSEMBLY: 9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Raw material	N/A	Dwg	Vendor lot certification
102	Hardness	N/A	Dwg	Vendor lot certification
103	Perpendicularity of aft side of flange with respect to cone axis of rotation	0.40%	Dwg	Gage
104	Cone surface straightness	0.40%	Dwg	Gage
105	Wall thickness and wall thickness variations	0.40%	Dwg	Gage
106	Runout of outside diameter of flange to the forward and aft diameter datums	0.40%	Dwg	Gage
107	Outside diameter	0.40%	Dwg	Gage
108	Diameter of cone at datum planes	0.40%	Dwg	Gage
109	Length of cone	0.40%	Dwg	Gage
Minor				
201	Workmanship	0.65%	3.3.2	4.5.10

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 16		DRAWING NUMBER
	RETAINER RING, LINER			9307005
				NEXT HIGHER ASSEMBLY
				9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Raw material	N/A	Dwg	Vendor lot certification
102	Hardness	N/A	Dwg	Vendor lot certification
Minor				
201	Hole locations and diameters	.65%	Dwg	Gage
202	Thread dimensions	.65%	Dwg	Gage
203	Diameter at Chamfer	.65%	Dwg	Gage
204	Length	.65%	Dwg	Gage
205	Workmanship	.65%	3.3.2	4.5.10

NOTES:

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: CONE, WARHEAD FREE SPACE	SHEET 3 OF 16		DRAWING NUMBER: 9307006
				NEXT HIGHER ASSEMBLY: 9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor				
201	Raw material	N/A	Dwg	Vendor lot certification
202	Flange O.D.	0.65%	Dwg	Gage
203	Notch size	0.65%	Dwg	Gage
204	Cone length	0.65%	Dwg	Gage
205	Notch location	0.65%	Dwg	Gage
206	Workmanship	0.65%	3.3.2	4.5.10
207	True position of cone surface with respect to O.D.	0.65%	Dwg	Gage
208	Flatness of flange	0.65%	Dwg	Gage
209	Size of tooling holes	0.65%	Dwg	Gage
210	Location of tooling holes	0.65%	Dwg	Gage

NOTES:

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 4 OF 16		DRAWING NUMBER
	END CLOSURE, WARHEAD			9307008
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	9307000
				PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	Thickness	100%	Dwg	Gage
2	Material	N/A	Dwg	Vendor lot certification
3	Hardness	100%	Dwg	Gage
Major	None defined			
Minor				
201	Thread dimensions	0.65%	Dwg	Gage
202	Workmanship	0.65%	3.3.2	4.5.10
203	Diameter of holes	0.65%	Dwg	Gage
204	Hole spacing	0.65%	Dwg	Gage

NOTES:

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: END CLOSURE, SEAL	SHEET 5 OF 16		DRAWING NUMBER: 9307009
				NEXT HIGHER ASSEMBLY: 9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor				
201	Raw material	N/A	Dwg	Vendor lot certification
202	Heat treatment	0.65%	Dwg	Gage
203	Inside diameter	0.65%	Dwg	Gage
204	Outside diameter	0.65%	Dwg	Gage
205	Thickness	0.65%	Dwg	Gage
206	Height of protrusion	0.65%	Dwg	Gage
207	Width of protrusion (minor)	0.65%	Dwg	Gage
208	Diameter of protrusion (major)	0.65%	Dwg	Gage
209	Workmanship	0.65%	3.3.2	4.5.10
210	True position of protrusion with respect to O.D.	0.65%	Dwg	Gage

NOTES:

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: RETAINER, FUZE	SHEET 6 OF 16		DRAWING NUMBER: 9307025
				NEXT HIGHER ASSEMBLY: 9307000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Material	N/A	Dwg	Vendor lot certification
102	Thickness	0.40%	Dwg	Gage
Minor				
201	Hole diameter	0.65%	Dwg	Gage
202	Hole location	0.65%	Dwg	Gage
203	Thread dimensions	0.65%	Dwg	Gage
204	Workmanship	0.65%	3.3.2	4.5.10

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 7 OF 16		DRAWING NUMBER
	WIRING HARNESS, WARHEAD-3W1-3W2			9307046
				NEXT HIGHER ASSEMBLY
				9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Wire resistance	0.40%	Dwg	Gage
Minor				
201	Cable length	0.65%	Dwg	Gage
202	Relief radius (two places)	0.65%	Dwg	Visual
203	Cable cutout (right end)	0.65%	Dwg	Gage
204	Cable slit cutout dimensions	0.65%	Dwg	Gage
205	Workmanship	0.65%	3.3.2	4.5.10
206	Tape installation	0.65%	Dwg	Visual

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: AFT CLOSURE, WARHEAD FORGING	SHEET 8 OF 16		DRAWING NUMBER: 9307051
				NEXT HIGHER ASSEMBLY: 9307001
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Material	N/A	Dwg	Vendor lot certification
Minor				
201	Workmanship	0.65%	3.3.2	4.5.10

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CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BODY, WARHEAD	SHEET 9 OF 16		DRAWING NUMBER: 9307001
				NEXT HIGHER ASSEMBLY: 9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	Inspection of weld and heat treatment, including cracks	100%	Dwg	9300967
2	Material	N/A	Dwg	Vendor lot certification
3	Dry lube on groove surfaces	100%	Dwg	Visual
4	Runout of aft interface shoulder to forward and aft diameter datums	100%	Dwg	Gage
5	Length of aft interface cylinder	100%	Dwg	Gage
6	Location of lock ring groove from aft interface shoulder	100%	Dwg	Gage
Major				
101	Runout of forward face to forward and aft diameter datums	0.40%	Dwg	Gage
102	Runout of ID in central portion of body to forward and aft diameter datums	0.40%	Dwg	Gage
103	Runout of OD at aft end of maximum diameter to forward and aft diameter datums	0.40%	Dwg	Gage

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE BODY, WARHEAD	SHEET 10 OF 16		DRAWING NUMBER 9307001
				NEXT HIGHER ASSEMBLY 9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
104	Runout of aft face to forward and aft diameter datums	0.40%	Dwg	Gage
105	True position of fuze well diameter	0.40%	Dwg	Gage
106	True position of thread at forward end	0.40%	Dwg	Gage
107	Liner centering diameter	0.40%	Dwg	Gage
108	True position of liner centering diameter to body ID	0.40%	Dwg	Gage
109	Runout of liner mounting shoulder to body ID	0.40%	Dwg	Gage
110	Diameter of fuze well	0.40%	Dwg	Gage
Minor				
201	Part number	0.65%	Dwg	Visual
202	Forward ID	0.65%	Dwg	Gage
203	Length of forward cable notch	0.65%	Dwg	Gage
204	Location of screw holes from forward end	0.65%	Dwg	Gage
205	Location of aft interface shoulder from forward end	0.65%	Dwg	Gage
206	Overall length	0.65%	Dwg	Gage

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 11 OF 16		DRAWING NUMBER
	BODY, WARHEAD			9307001
				NEXT HIGHER ASSEMBLY
				9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
207	Width of lock ring groove	0.65%	Dwg	Gage
208	Diameter of lock ring groove	0.65%	Dwg	Gage
209	Diameter of aft interface cylinder	0.65%	Dwg	Gage
210	Maximum diameter	0.65%	Dwg	Gage
211	Length of forward alignment pin slot	0.65%	Dwg	Gage
212	Width of forward alignment pin slot	0.65%	Dwg	Gage
213	True position of forward alignment pin slot	0.65%	Dwg	Gage
214	Location of fuze well keyway from body centerline	0.65%	Dwg	Gage
215	Width of fuze well keyway	0.65%	Dwg	Gage
216	True position of fuze well keyway	0.65%	Dwg	Gage

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BODY, WARHEAD	SHEET 12 OF 16		DRAWING NUMBER: 9307001
				NEXT HIGHER ASSEMBLY: 930/011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
217	Diameter of forward interface screwholes	0.65%	Dwg	Gage
218	True position of screwholes (16) in forward interface	0.65%	Dwg	Gage
219	Thread dimensions at forward end	0.65%	Dwg	Gage
220	Location of fuze well keyway from longitudinal centerline	0.65%	Dwg	Gage
221	Location of liner mounting shoulder from forward face	0.65%	Dwg	Gage
222	Length of liner centering diameter	0.65%	Dwg	Gage
223	Diameter of liner mounting shoulder ID	0.65%	Dwg	Gage
224	True position of liner mounting shoulder ID to liner centering diameter	0.65%	Dwg	Gage
225	Length of cable penetration slot	0.65%	Dwg	Gage
226	Thread dimensions at aft end	0.65%	Dwg	Gage
227	True position of thread (aft end to fuze well ID)	0.65%	Dwg	Gage

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 13 OF 16		DRAWING NUMBER
	BODY, WARHEAD			9307001
				NEXT HIGHER ASSEMBLY
				9307011
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
228	Location of torque lock notch from forward face	0.65%	Dwg	Gage
229	Length of thread, aft end	0.65%	Dwg	Gage
230	Perpendicularity of aft thread to aft face	0.65%	Dwg	Gage
231	Width of torque lock notches (2)	0.65%	Dwg	Gage
232	True position of torque lock notches (2)	0.65%	Dwg	Gage
233	Workmanship	0.65%	3.3.2	4.5.10

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: WARHEAD, EMPTY, GUIDED PROJECTILE	SHEET 14 OF 16		DRAWING NUMBER 9307011
				NEXT HIGHER ASSEMBLY 9307055
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	No unsolidified paint in charge cavity	100%	3.2.1.7	4.5.9
2	Finished surfaces	100%	3.2.1.7	4.5.9
3	Adhesive	N/A	Spec	9300922 - Vendor lot certification
Major				
101	Electrical harness installation	0.40%	Dwg	Visual and gage
102	Electrical harness characteristics	100%	3.2.1.1	4.5.4.1
103	Liner and cone installation	0.40%	Dwg	Visual
Minor				
201	Workmanship	0.65%	3.3.2	4.5.10
202	Retainer ring torque	0.65%	Dwg	Gage
203	Marking (including serialization)	0.65%	Dwg	Visual
204	Max O.D. at warhead, aft end in area of wrap	0.65%	Dwg	Gage
205	Filler	N/A	Spec	MIL-S-14760 - Vendor lot certification
206	Leak Test	0.65%	3.2.1.4	4.5.3

NOTES:

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE WARHEAD, LOADED, GUIDED PROJECTILE (AT AMMUNITION LOADING PLANT)	SHEET 15 OF 16		DRAWING NUMBER 9307055 NEXT HIGHER ASSEMBLY 9307000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	No explosive material, other than within charge cavity	100%	3.2.1.6	4.5.9
2	No loose explosive material in the fuze well	100%	3.2.1.7	4.5.9
3	Defects in explosive charge	100%	3.2.1.5.7	1 4.5.5, 4.5.6, 4.5.9
Major				
101	Electrical harness characteristics (see Note 1)	100%	3.2.1.2	4.5.4.2
102	Defects in explosive charge	100%	3.2.1.5.7	2 4.5.5
Minor				
201	Warhead lot number	0.65%	Dwg	Visual
202	Workmanship	0.65%	3.3.2	4.5.10
203	Fuze well depth	0.65%	Dwg	Gage
204	Fuze well diameter	0.65%	Dwg	Gage
205	Location of fuze well within metal body ID	0.65%	Dwg	Gage
206	Cable covers properly installed	100%	Dwg	Visual

NOTES: 1 - This inspection to be performed after return from loading plant.

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Table V
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 16 OF 16		DRAWING NUMBER 9307000
	WARHEAD, GUIDED PROJECTILE			NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	Green Flag - Safe indicators (at installation of fuze)	100%	Dwg	Visual
2	Fuze shock pad installation	100%	Dwg	Visual
Major				
101	Warhead section wiring	100%	3.2.1.3	4.5.4.3
102	Fuze cable installation	0.40%	Dwg	Visual
103	End closure seal installation	0.40%	Dwg	Visual
Minor				
201	Examination for damage due to handling and transportation	0.65%	3.3.2	4.5.10
202	Fuze retainer torqued	0.65%	Dwg	Gage
203	End closure torqued	0.65%	Dwg	Gage
204	Workmanship	0.65%	3.3.2	4.5.10
205	Preparation for delivery	0.65%	5.0	4.5.11

NOTES:

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Table VI
CONFORMANCE INSPECTIONS, LOT ACCEPTANCE
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE WARHEAD, GUIDED PROJECTILE (AT AMMUNITION LOADING PLANT - SEE NOTE 1)	SHEET 1 OF 1		DRAWING NUMBER 9307055 NEXT HIGHER ASSEMBLY 9307000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical				
1	No explosive material other than within change cavity	100%	3.2.1.6	4.5.9
2	No loose explosive material in the fuze well	100%	3.2.1.7	4.5.9
3	Defects in explosive charge	100%	3.2.1.5.7	4.5.5, 4.5.6, 4.5.9, review existing radiographs
Major				
101	Specific gravity of explosive	100%	Dwg	Gage
102	Static penetration	100%	3.2.1.8	4.5.8
Minor				
201	Warhead lot number	100%	Dwg	Visual
202	Workmanship	100%	3.3.2	4.5.10

NOTES: 1 - Two units shall be selected at random from each sublot for this test. Sublot is defined in 4.4.2.2.

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